

Weekly Challenge 02: Deterministic Finite Automata

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1. For a language $L \subseteq \Sigma^*$, the $\neg$ operator is defined as follows: $\neg(L) = \{w \in \Sigma^* \mid w \notin L\}$.

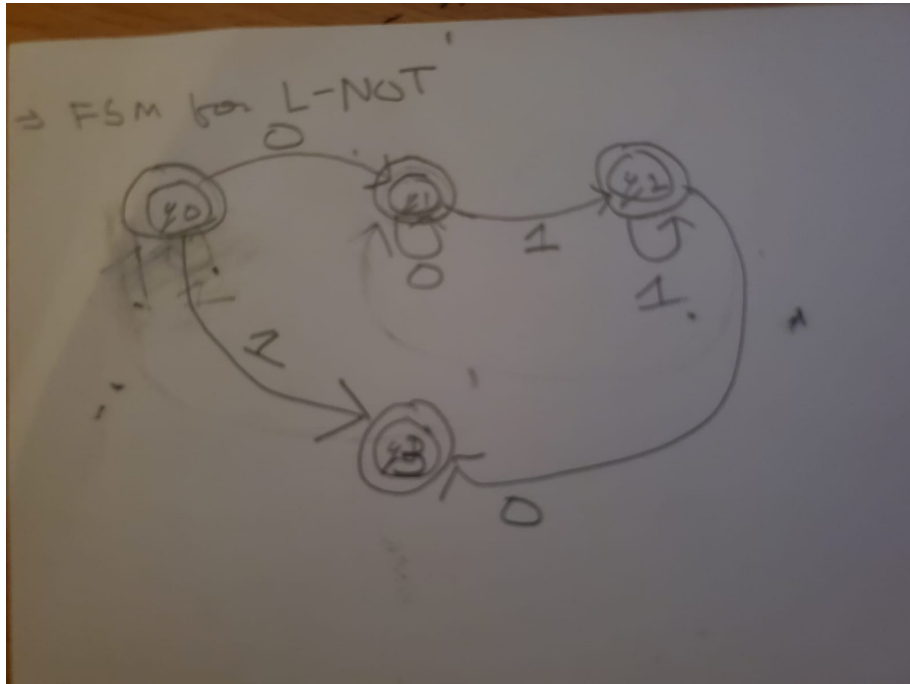
Consider the language L of strings of form $0^n 1^m$ for $n, m \in \mathbb{Z}^+$.

- (a) Define $\neg(L)$, and construct a FSM (DFA) for $\neg(L)$.

Solution:

$\neg(L) = \{w \in \Sigma^* \mid w \in 0, 1^* \text{ w not in form } 0^n 1^m \text{ for } n, m \text{ in } \mathbb{Z}\}$

we can draw the DFA for L , then invert the accepting states to get $\neg(L)$



DFA

- (b) Prove or disprove the following claim: If a language $L \subseteq \Sigma^*$ is regular then $\neg(L)$ is also regular.

Solution:

If L is a regular DFA defined $(Q, \Sigma, \delta, q_0, F)$, then there exists another DFA L' such:
 $L' = (Q, \Sigma, \delta, q_0, Q \setminus F)$

This state accepts all states that are complement of L , as its accepting states are the not in set F of the accepting states of L with other definitions identical.
Therefore, $w \in L' \iff \delta(q_0, w) \in Q \setminus F$